Lab 07: Statistical Analysis of Housing Dataset

Cell 1: Import libraries and load the dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the existing dataset
df = pd.read_csv('Housing.csv')

# Display the first five rows
print("FIRST FIVE ROWS OF THE DATASET:")
display(df.head())

# Display dataset information
print("\nDATASET INFORMATION:")
df.info()

# Display dataset shape
print("\nDATASET SHAPE:", df.shape)
```

```
FIRST FIVE ROWS OF THE DATASET:
```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	pr
0	13300000	7420	4	2	3	yes	no	no	no	yes	2	
1	12250000	8960	4	4	4	yes	no	no	no	yes	3	
2	12250000	9960	3	2	2	yes	no	yes	no	no	2	
3	12215000	7500	4	2	2	yes	no	yes	no	yes	3	
4	11410000	7420	4	1	2	yes	yes	yes	no	yes	2	

```
DATASET INFORMATION:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 545 entries, 0 to 544
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   price             545 non-null    int64
 1   area              545 non-null    int64
 2   bedrooms          545 non-null    int64
 3   bathrooms         545 non-null    int64
 4   stories           545 non-null    int64
 5   mainroad          545 non-null    object
 6   guestroom         545 non-null    object
 7   basement          545 non-null    object
 8   hotwaterheating   545 non-null    object
 9   airconditioning   545 non-null    object
 10  parking           545 non-null    int64
 11  prefarea          545 non-null    object
 12  furnishingstatus  545 non-null    object
dtypes: int64(6), object(7)
memory usage: 55.5+ KB

DATASET SHAPE: (545, 13)
```

Cell 2: Calculate mean, median, mode, and standard deviation

```
# Select numerical columns
numeric_df = df.select_dtypes(include=np.number)

# Calculate mean
mean_values = numeric_df.mean()

# Calculate median
median_values = numeric_df.median()

# Calculate mode for all columns
mode_values = df.mode().iloc[0]

# Calculate standard deviation
std_values = numeric_df.std()
```

```
# Display results
print("MEAN:")
print(mean_values)

print("\nMEDIAN:")
print(median_values)

print("\nMODE:")
print(mode_values)

print("\nSTANDARD DEVIATION:")
print(std_values)
```

```
MEAN:
price        4.766729e+06
area         5.150541e+03
bedrooms     2.965138e+00
bathrooms    1.286239e+00
stories      1.805505e+00
parking      6.935780e-01
dtype: float64

MEDIAN:
price        4340000.0
area            4600.0
bedrooms           3.0
bathrooms          1.0
stories            2.0
parking            0.0
dtype: float64

MODE:
price                     3500000
area                       6000.0
bedrooms                      3.0
bathrooms                     1.0
stories                       2.0
mainroad                      yes
guestroom                      no
basement                       no
hotwaterheating                no
airconditioning                no
parking                       0.0
prefarea                       no
furnishingstatus    semi-furnished
Name: 0, dtype: object

STANDARD DEVIATION:
price        1.870440e+06
area         2.170141e+03
bedrooms     7.380639e-01
bathrooms    5.024696e-01
stories      8.674925e-01
parking      8.615858e-01
dtype: float64
```

Cell 3: Calculate and display the correlation matrix

```
# Convert categorical columns into numerical dummy variables
encoded_df = pd.get_dummies(df, drop_first=True, dtype=int)

# Calculate correlation matrix
correlation_matrix = encoded_df.corr()

# Display correlation matrix
print("CORRELATION MATRIX:")
display(correlation_matrix.round(2))
```

```
CORRELATION MATRIX:
```

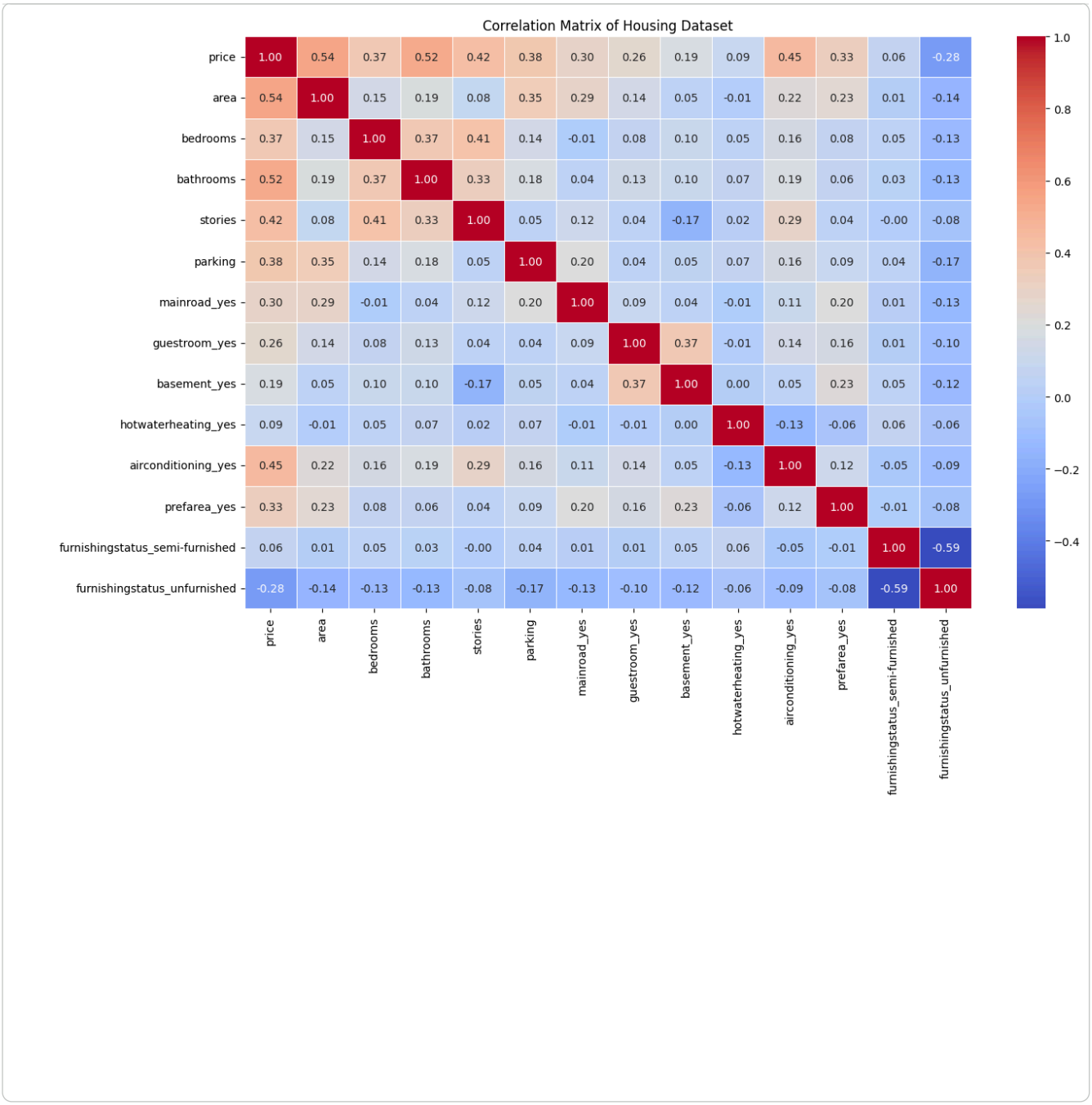
	price	area	bedrooms	bathrooms	stories	parking	mainroad_yes	guestroom_yes	basement_yes	ho
price	1.00	0.54	0.37	0.52	0.42	0.38	0.30	0.26	0.19	
area	0.54	1.00	0.15	0.19	0.08	0.35	0.29	0.14	0.05	
bedrooms	0.37	0.15	1.00	0.37	0.41	0.14	-0.01	0.08	0.10	
bathrooms	0.52	0.19	0.37	1.00	0.33	0.18	0.04	0.13	0.10	
stories	0.42	0.08	0.41	0.33	1.00	0.05	0.12	0.04	-0.17	
parking	0.38	0.35	0.14	0.18	0.05	1.00	0.20	0.04	0.05	
mainroad_yes	0.30	0.29	-0.01	0.04	0.12	0.20	1.00	0.09	0.04	
guestroom_yes	0.26	0.14	0.08	0.13	0.04	0.04	0.09	1.00	0.37	
basement_yes	0.19	0.05	0.10	0.10	-0.17	0.05	0.04	0.37	1.00	
hotwaterheating_yes	0.09	-0.01	0.05	0.07	0.02	0.07	-0.01	-0.01	0.00	
airconditioning_yes	0.45	0.22	0.16	0.19	0.29	0.16	0.11	0.14	0.05	
prefarea_yes	0.33	0.23	0.08	0.06	0.04	0.09	0.20	0.16	0.23	
furnishingstatus_semi-furnished	0.06	0.01	0.05	0.03	-0.00	0.04	0.01	0.01	0.05	
furnishingstatus_unfurnished	-0.28	-0.14	-0.13	-0.13	-0.08	-0.17	-0.13	-0.10	-0.12	

Cell 4: Plot the correlation matrix

```
# Plot correlation heatmap
plt.figure(figsize=(14, 10))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap='coolwarm',
    fmt='.2f',
    linewidths=0.5
)

plt.title("Correlation Matrix of Housing Dataset")
plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
```



Cell 5: Calculate and display the covariance matrix

```
# Calculate covariance matrix
covariance_matrix = encoded_df.cov()

# Display covariance matrix
print("COVARIANCE MATRIX:")
display(covariance_matrix.round(2))
```

```
COVARIANCE MATRIX:
```

	price	area	bedrooms	bathrooms	stories	parking	mainroad_yes	guestroom_y
price	3.498544e+12	2.175676e+09	505946.43	486409.33	682644.63	619467.32	193607.54	182974.
area	2.175676e+09	4.709512e+06	243.23	211.35	158.13	659.99	218.56	116.
bedrooms	5.059464e+05	2.432300e+02	0.54	0.14	0.26	0.09	-0.00	0.
bathrooms	4.864093e+05	2.113500e+02	0.14	0.25	0.14	0.08	0.01	0.
stories	6.826446e+05	1.581300e+02	0.26	0.14	0.75	0.03	0.04	0.
parking	6.194673e+05	6.599900e+02	0.09	0.08	0.03	0.74	0.06	0.
mainroad_yes	1.936075e+05	2.185600e+02	-0.00	0.01	0.04	0.06	0.12	0.
guestroom_yes	1.829748e+05	1.165600e+02	0.02	0.02	0.01	0.01	0.01	0.
basement_yes	1.670850e+05	4.914000e+01	0.03	0.02	-0.07	0.02	0.01	0.
hotwaterheating_yes	3.645362e+04	-4.190000e+00	0.01	0.01	0.00	0.01	-0.00	-0.
airconditioning_yes	3.941112e+05	2.245100e+02	0.06	0.04	0.12	0.06	0.02	0.
prefarea_yes	2.617215e+05	2.161800e+02	0.02	0.01	0.02	0.03	0.03	0.
furnishingstatus_semi-furnished	5.875085e+04	6.590000e+00	0.02	0.01	-0.00	0.02	0.00	0.
furnishingstatus_unfurnished	-2.463526e+05	-1.449300e+02	-0.04	-0.03	-0.03	-0.07	-0.02	-0.

Cell 6: Plot the covariance matrix

```
# Plot covariance heatmap
plt.figure(figsize=(12, 9))

off_diagonal = covariance_matrix.values[
    ~np.eye(covariance_matrix.shape[0], dtype=bool)
]

limit = np.percentile(np.abs(off_diagonal), 95)

sns.heatmap(
    covariance_matrix,
    cmap='coolwarm',
    center=0,
    vmin=-limit,
    vmax=limit,
    annot=True,
    fmt='.2e',
    annot_kws={"size": 7},
    linewidths=0.3,
    cbar_kws={"shrink": 0.7}
)

plt.title("Covariance Matrix of Housing Dataset", fontsize=12)
plt.xticks(rotation=90, fontsize=8)
plt.yticks(rotation=0, fontsize=8)
plt.tight_layout()
plt.show()
```


Covariance Matrix of Housing Dataset